

Zineb ZIANI

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EXPERIENCE

ANOTHERBRAIN | AI RESEARCH ENGINEER

Feb 2025 – Currently | Paris

- Designing a cortical-inspired module for real intelligence, based on the structure of cortical columns.
- Conducting research on a memory-based recognition system inspired by biological mechanisms, using image descriptors.

RIKEN CENTER FOR COMPUTATIONAL SCIENCE

VISITING RESEARCHER | AI AND HPC ENGINEER

July 2024 – Oct 2024 | Tokyo - Japan

- Leveraging Parallel Deep Learning and Spectral Methods, including RNNs, GCNs, and MLP, for Advanced Parallel Anomaly Detection on the Fugaku Supercomputer.

RIKEN CENTER FOR COMPUTATIONAL SCIENCE

VISITING RESEARCHER | AI AND HPC ENGINEER

June 2023 – Sept 2023 | Kobe - Japan

- Optimized AI anomaly detection framework on the supercomputer Fugaku, leveraging Python and gaining global-scale insights in HPC.

CEA / NUMERYX TECHNOLOGIES | AI AND HPC ENGINEER

Dec 2021 – Jan 2025 | Paris

- Deploying anomaly detection solutions by combining machine learning and deep learning models with the "Unite and Conquer" HPC strategy, leveraging parallel processing on large datasets to enhance speed and accuracy.
- Integrating this framework into a SOAR system and deploying it into production.

ORANGE | MACHINE LEARNING ENGINEER

Feb 2021 – Aug 2021 | Caen

- Study and evaluation of training methods allowing to reduce the size of a model, like LSTM and BERT while at the same time maintaining its statistical efficiency, by addressing "transfer learning", "distillation", "quantization" among others.

MICROLIGHT 3D | 3D GEOMETRIC MODELLING ENGINEER

May 2020 – Aug 2020 | Grenoble

- implement a new "slicing" method in order to limit the manufacturing volume and bring consistency with mechanical strength by using geometry libraries in C++.

PUBLICATIONS

Enhancing the Parallel UC2B Framework: Approach Validation and Scalability Study

24th International Conference on Computational Science; Springer 2024

Unite and Conquer with Bagging-Boosting "UC2B" for Enhanced Anomaly Detection

2024 IEEE International Conference on Progress in Informatics Computing

A Parallel and Asynchronous Approach for Anomaly Detection

2024 IEEE International Conference on Big Data

A Novel Approach to Parallel Anomaly Detection: Application in

Cybersecurity 2023 IEEE International Conference on Big Data

EDUCATION

PhD in Mathematics Informatics

AI and HPC for enhanced anomaly detection Feb 2022 - Jan 2025 |

University of Paris-Saclay, Paris

- Enhancing threat detection through data science, parallel computing, and supercomputers for more effective incident management.

MSc In Industrial an Applied Mathematics | Data Science

2019 - 2021 | UGA-ENSIMAG, Grenoble

- Advanced learning Models
- Machine learning • NLP
- fundamentals of probabilistic data mining • Statistical analysis and document mining • Category learning and object recognition • Computing science for Big Data and HPC
- Applied probability and statistics.

Licence de Mathématiques Fondamentales

2017 - 2019 | University of Grenoble-Alpes, Grenoble

- Topology • Statistics • Algebra
- Numerical modeling • PDE
- Integral calculus & probability.

SKILLS

- Python and R Programming
- Machine & Deep Learning Frameworks
- Natural Language Processing (NLP): Autoencoders, Transformers
- Vulnerability Analysis & Detection
- High-Performance Computing: Supercomputers, Clusters, Parallelism
- Linux command-line operations, job management and monitoring
- SLURM and PJM for job scheduling and resource allocation
- Skilled in managing and optimizing HPC compute nodes and storage.

CONFERENCES

Committee Chair of IEEE BigData

Reviewer : ICCS 2023, ICCS 2024, ICCS 2025, IIAI AAI 2023, IIAI AAI 2024, IIAI AAI 2024-Winter

LANGUAGES

French, English, Arabic.